How We Label

The guide for the Bangla ASR labeling work: references, error audit, word pairs, and settling disagreements.

CSE499B · Omi, Nabigah, Nafisa Version 1 · 9 October 2026 Site: asr-labelling-kappa.vercel.app

1. Why this work matters

Our labels are the only part of the project that a computer cannot produce. They answer three questions the paper depends on:

- **Are the dataset's transcripts right?** If a transcript is wrong, the model gets blamed for an error it did not make. Your corrected references tell us how often that happens.
- **Are the "spelling errors" really spelling?** Our program guesses that about 40% of the model's errors are the same word written differently (ছিল / ছিলো). Your judgments tell us whether that guess is true. This decides the main claim of the paper (Gate 3).
- **Can other people trust our labels?** Some items are labeled by two or three of us without talking. How often we agree (called κ, "kappa") goes in the paper as proof that the labels are reliable.

> **Three things matter more than speed**
>
> 1. **Consistency:** all three of us follow exactly the same rules. A rule we apply differently makes our agreement look bad, even if each of us was "right".
> 2. **Independence:** never discuss an item before all three have labeled it. Disagreements are settled together at the end, on a separate screen.
> 3. **Honesty:** when you are not sure, say so (with a note or "Can't tell"). A guess that looks confident is worse than an honest "unclear".

Our labels are used only to **measure** results, never to train or tune a model. That is why they must reflect the audio, not what we think the model or the paper needs.

2. The work at a glance

Task	What you do	Items	Time per item	Done by
1. References	Listen to a clip and make the transcript match exactly what is said.	900 clips (600 DL Sprint, 100 each FLEURS, Kathbath, Common Voice)	2–3 min	one person; 10% by two people
2. Error audit	For every place where the transcript and the recognizer differ, judge who is right.	400 clips	1–2 min	one person; first 60 by all three
3. Word pairs	Decide what kind of difference two words are.	1,000 pairs	about 10 s	one person; first 150 by all three

In total that is about **2,800 labels, roughly 19 hours each**. Shared items are served first, so the agreement check can start early.

Order: References first, then the error audit, then word pairs. References come first because you should hear and fix the clips before you have seen what the recognizer wrote.

Sessions: work in blocks of 45–60 minutes with a break. Use headphones. Labeling quality drops quickly when you are tired; stop and come back.

3. Using the site

1. Open **asr-labelling-kappa.vercel.app**, enter the **team passcode** (ask Omi; never post it anywhere) and choose your name.
2. The home screen shows the three tasks with team progress and your own count, the **Settle disagreements** buttons, and a table of everyone's progress.
3. Press **Start** on a task. The first time, its guide opens on top; you can reopen it any time with the **Guide** button in the header.
4. Label the item and press **Save & next**. Everything saves to the shared database at that moment.
5. To fix an earlier answer, press ← **Previous** or **My finished items**, open the item, change it and save again.

Key	Does
Ctrl + Space	Play / pause the clip (all tasks)
Ctrl + Enter	Save and go to the next item
1 – 6	Error audit: label the selected difference and move to the next one
1 – 4	Word pairs: label the pair, save and move on

References keep an unsaved draft in your browser, so a closed tab does not lose your typing. Only "Save & next" puts it in the shared database.

4. The golden rules

Almost every hard case is answered by one of these. When unsure, come back to this page.

1. **Write what you hear, not what the context suggests.** If the speaker clearly says ভাবে where ভাবি is meant, write ভাবে and add the note `means ভাবি`. Never "fix" a word because the sentence would make more sense.
2. **Fix words, never punctuation.** Ignore । , ? ! quotes and the `>>` speaker marks, even when they are wrong. Scoring removes them anyway. If only punctuation is off, the clip is **Correct as is**.
3. **Do not change a valid spelling.** ছিল / ছিলো, কি / কী, কোন / কোনো, হল / হলো: if the spoken word is written in any accepted spelling, leave it.
4. **Dialect word forms are written as spoken.** Speaker says কইরা and the text has করে: change it to কইরা. Same for নিয়া, তুমার, যাইতাছি. If the speaker uses the standard form, keep it.
5. **An accent on one sound keeps the standard spelling.** Speaker says ঢেকালো for ঠেকালো, or "bhuk" for ভুখ: write the standard spelling and add the note `accent`. The difference between rules 4 and 5: in a dialect form the *word* changes (করে → কইরা); in an accent one *sound* is said differently but it is the same written word.
6. **Do not guess.** If you cannot make out a word after listening several times, keep the original word and add the note `unclear`. If you cannot make out the main speech at all, choose **Can't use**.
7. **Work alone on shared items.** Do not ask each other about an item until it is in "Settle disagreements".

5. Task 1: References

You see the clip, the **Original transcript**, and a box **What is actually said** that starts as a copy of it. Edit the box until it matches the audio, then choose one status:

Status	Choose it when
Correct as is	Every word matches the audio (punctuation does not matter).
Corrected	You changed at least one word. The site picks this for you as soon as you change a word.
Can't use	The main speech cannot be made out: music, noise, people talking over each other, or another language.

Tick **Words cut off at the start or end** when the clip starts or ends in the middle of a word. Keep that word if more than half of it is audible; otherwise remove it.

What to do in each situation

Situation	Do this	Example
A word is missing from the transcript	Add it.	আমি খাই → আমি ভাত খাই
The transcript has a word that is not spoken	Remove it.	

A wrong word	Replace it with the spoken word.	তোমার কি ভোগ লেগেছে → তোমার কি ভুখ লেগেছে
Accepted spelling variant	Leave it.	ছিলো stays ছিলো
Dialect form spoken	Write the dialect form.	করে → কইরা
Accent on one sound	Standard spelling + note `accent`.	ঢেকালো → ঠেকালো
Clearly says a word, context suggests another	Write what is said + note `means …`.	ভাবে + note "means ভাবি"
A word you cannot make out	Keep the original + note `unclear`.	
Numbers	As words, as spoken.	২৫ → পঁচিশ
English words	In Bangla script, as the transcript does.	ফোন, অফিস
Fillers and hesitations	Leave them out.	উম, আ…
A word said twice on purpose	Write it twice.	গরম গরম
A false start or stutter	Leave out the broken part.	আ- আমি → আমি
Background speech, laughter, music	Only the main speaker; no tags like [মিউজিক].	
Names	Keep the transcript's spelling unless it is clearly a different name.	

6. Task 2: Error audit

You see the clip, the **transcript** from the dataset, and a **recognizer's output**. Every place where they differ is highlighted. Select a difference (or let the site move through them) and press a key:

Key	Label	Choose it when	Example
1	**Recognizer wrong**	The transcript matches the audio, the recognizer does not.	
2	**Transcript wrong**	The recognizer matches the audio, the transcript does not.	transcript ভোগ, recognizer ভুখ, speaker says ভুখ
3	**Both wrong**	Neither matches what is said.	
4	**Same word, spelling differs**	Both write the spoken word correctly, in different accepted spellings, or split differently.	ছিল / ছিলো, কি / কী, দু পাশে / দুপাশে
5	**Same word, dialect vs standard**	One side writes the spoken dialect form, the other the standard form. Also an accent on one sound.	করে / কইরা, তোমার / তুমার, ঠেকালো / ঢেকালো
6	**Can't tell**	The audio is unclear at this spot.	

How to decide, in order

1 Listen to the spot. Judge against the **audio**, never against which version looks nicer or makes more sense.

2 Is it the same word on both sides? If yes: two accepted spellings → 4; a dialect form or an accent → 5.

3 If not the same word: whose word matches the audio? Transcript → 1; recognizer → 2; neither → 3.

4 Still unsure after listening again → 6. Do not guess.

Common traps

- A missing or extra word is not a spelling difference. If the recognizer left out a spoken word, that is **1**; if the transcript has a word nobody says, that is **2**.
- আর / আরো, এখন / এখনো, করে / করা are **different words** (meaning or grammar changes), so they are 1, 2 or 3, not 4.
- When the speaker clearly says the recognizer's word but context suggests the transcript's, the transcript is wrong: **2** (golden rule 1).

7. Task 3: Word pairs

You see a word from a transcript (left) and the word a recognizer wrote instead (right), with two example sentences. Decide what kind of difference it is; the key saves and moves on.

Key	Label	Meaning	Examples
1	**Same word, different spelling**	Same word and meaning, written two accepted ways.	ছিল / ছিলো, কি / কী, রং / রঙ, কোন / কোনো
2	**Dialect or colloquial form**	The same word said the regional or spoken way, including an accent on one sound.	নিয়ে / নিয়া, তোমার / তুমার, ঠেকালো / ঢেকালো
3	**Different word**	The meaning or grammar changes.	আর / আরো, এখন / এখনো, করে / করা
4	**Can't tell**	The examples do not make it clear.	

Judge the two **words**, using the example sentences only for context. There is no audio in this task: you are not judging who was right, only what kind of difference it is.

8. Hard cases (questions we have already met)

Question	Answer
I can't understand one word, even after several listens.	Keep the transcript's word, note `unclear`, label the rest normally. Audit: 6 for that spot.

I can't understand the clip at all.	**Can't use**. That is a valid answer: it tells us how much of the test data is unusable.
The speaker clearly says ভাবি; the transcript has ভাবে.	Change it to ভাবি. You heard it.
The speaker clearly says ভাবে, but the sentence means ভাবি.	Write ভাবে, note `means ভাবি`. Sound wins over context.
The speaker says ঢেকালো for ঠেকালো.	Accent on one sound: keep ঠেকালো, note `accent`.
I corrected a wrong word but am not sure how to spell the right one.	Use the standard dictionary spelling of the spoken word (hunger is ভুখ, not ভুক). If you are not sure it is the right word at all, keep the original and note `unclear`.
The transcript writes a dialect word in standard form, and the speaker uses the dialect form.	Write the dialect form (golden rule 4).
Only the punctuation or the `>>` marks are wrong.	**Correct as is**.
Two people talk at once.	Transcribe the main speaker. If you cannot tell who the main speaker is, **Can't use**.
A song or heavy music covers the speech.	Transcribe the speech if it is clear; otherwise **Can't use**.
I made a mistake on an item I already saved.	**My finished items** → open it → fix → save again.
My answer on a shared item differs from a teammate's.	Do nothing alone. It appears in "Settle disagreements" for all of us.

The note box: use these words

Notes are counted by a program later, so use the same words every time. You can add a few words after them.

Write	When
`unclear`	A word you could not make out (add which one: `unclear: ঠেকালো`).
`accent`	You wrote the standard spelling of a word said with an accent.
`means …`	The speaker said one word but clearly meant another (`means ভাবি`).
`music`, `noise`, `overlap`, `other language`	Why a clip is **Can't use**.

9. Settling disagreements

When everyone has finished a shared item and the answers differ, it appears under **Settle disagreements** on the home screen. Do this **together** (in person or on a call), after your separate sessions.

1. Open the task's button (it shows how many are open). Items come one by one.
2. **References** show what each person changed. In the original transcript, words someone changed are highlighted. In each answer, removed words are ~~struck through~~ and new words are green, with a summary line such as "Changes: ভাবে → ভাবি · ভোগ → ভুখ". So you only listen to the spots that differ.
3. Listen to those spots together and apply the golden rules. Press **Use this text** on the closest answer, edit the **Agreed text** if needed, choose the status and press **Save agreed answer**.
4. **Error audit** shows every difference with each person's label; only the ones that differ need an agreed choice. **Word pairs** show everyone's label; pick the agreed one.
5. If, after listening together, you still cannot agree, use the honest answer: the original word with `unclear` (references) or **Can't tell** (audit, pairs).

Your original answers are never changed: the agreed answer is stored separately. Agreement (κ) is computed from the original, independent answers, so it is important that those were made alone.

Quick card

Write what you hear · words, not punctuation · valid spellings stay · dialect forms as spoken · accent → standard spelling + `accent` · sound beats context (`means …`) · unsure → original + `unclear` · nothing audible → **Can't use** · shared items: alone first, together at the end.
Ctrl + Space play · Ctrl + Enter save & next · audit 1–6 · pairs 1–4

Questions about a rule? Ask in the group chat **about the rule, never about a specific shared item**. If we add or change a rule, this guide gets a new version number and the site's guide is updated the same day.